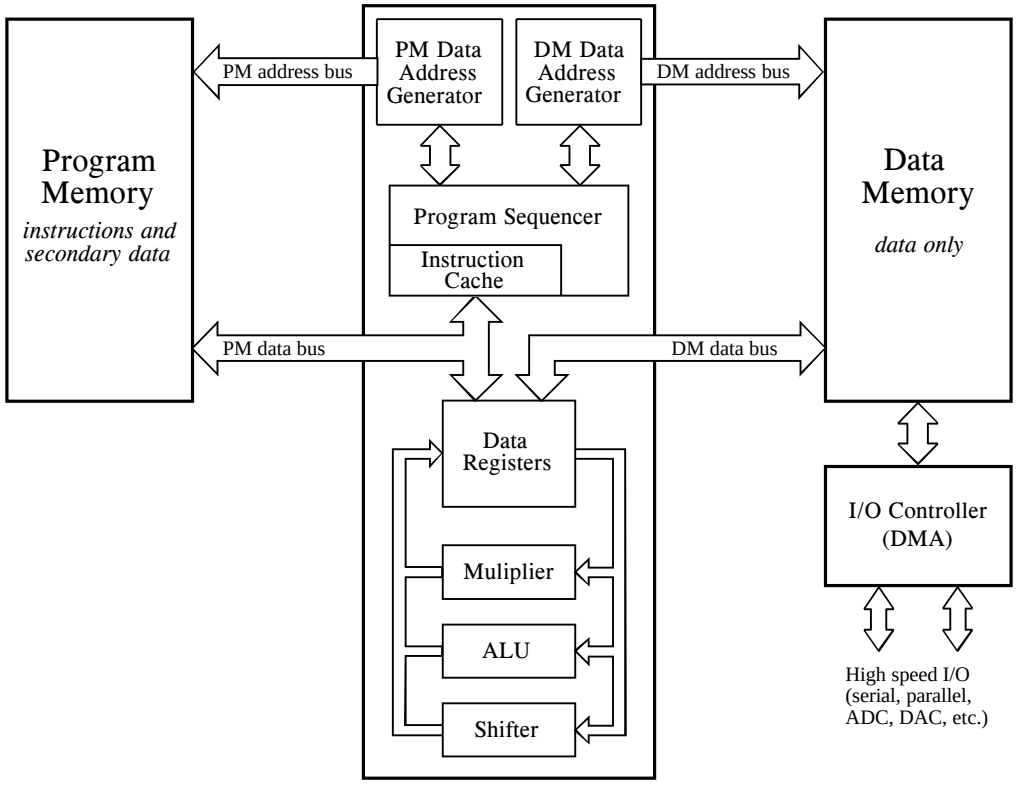

FIGURE 28-5
Typical DSP architecture. Digital Signal Processors are designed to implement tasks in parallel. This simplified diagram is of the Analog Devices SHARC DSP. Compare this architecture with the tasks needed to implement an FIR filter, as listed in Table 28-1. All of the steps within the loop can be executed in a single clock cycle.

feature is the use of **shadow registers** for all the CPU's key registers. These are duplicate registers that can be switched with their counterparts in a single clock cycle. They are used for *fast context switching*, the ability to handle interrupts quickly. When an interrupt occurs in traditional microprocessors, all the internal data must be saved before the interrupt can be handled. This usually involves pushing all of the occupied registers onto the stack, one at a time. In comparison, an interrupt in the SHARC family is handled by moving the internal data into the shadow registers in a *single clock cycle*. When the interrupt routine is completed, the registers are just as quickly restored. This feature allows step 4 on our list (managing the sample-ready interrupt) to be handled very quickly and efficiently.

Now we come to the critical performance of the architecture, how many of the operations within the loop (steps 6-12 of Table 28-1) can be carried out at the same time. Because of its highly parallel nature, the SHARC DSP can simultaneously carry out *all* of these tasks. Specifically, within a single clock cycle, it can perform a multiply (step 11), an addition (step 12), two data moves (steps 7 and 9), update two circular buffer pointers (steps 8 and 10), and